



Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL)

marp:true

These three terms are closely related, but they are not the same. Think of them as nested circles:

- **AI** is the biggest circle
- **ML** is a part of AI
- **DL** is a part of ML

1 Artificial Intelligence (AI)

Definition:

AI is the field of computer science that makes machines perform tasks that normally require human intelligence.

Examples of AI:

- Chatbots that answer customer questions
- Self-driving cars
- Voice assistants like Siri or Alexa

Key Idea:

AI is about making intelligent machines, not just software that follows strict rules.

2 Machine Learning (ML)

Definition:

Machine Learning is a subset of AI that allows machines to learn from data and improve automatically without being explicitly programmed.

How it works:

1. The machine is given data (inputs and sometimes outputs).
2. It finds patterns in the data.
3. It makes predictions or decisions based on what it learned.

Examples of ML:

- Predicting house prices based on historical data
- Detecting spam emails
- Recommendation systems like Netflix or YouTube

3 Deep Learning (DL)

Definition:

Deep Learning is a subset of Machine Learning that uses artificial neural networks with many layers to solve complex problems.

Why "Deep"?

The word "deep" refers to the many layers in the neural network.

Examples of DL:

- Image recognition (e.g., detecting cats and dogs in photos)
- Self-driving cars recognizing traffic signs
- Speech recognition and translation

Key Idea:

DL is best for large, complex datasets like images, video, and audio.

◆ Comparison Table

Feature	AI	ML	DL
Goal	Simulate human intelligence	Learn from data	Learn from large complex data
Data Needed	Not always	Medium	Very large datasets
Programming	Rules-based or learning-based	Learning-based	Neural networks
Examples	Chess-playing AI, Robots	Spam detection, Recommendations	Image recognition, Voice assistants

Simple Way to Explain

- **AI** → The big goal: machines think
- **ML** → Machines learn from data
- **DL** → Machines learn complex patterns using neural networks